TEXORA · LECTURE SHEET

Fabric Stiffness, Handle and Drape

Strength tells us whether a fabric will survive; stiffness, handle and drape tell us whether anyone will want to wear it. This sheet covers cantilever bending and Peirce flexural rigidity, drape and the Cusick drape coefficient, and handle as the sum of the mechanical properties the hand can feel.

TE-3201 · QUALITY EVALUATION OF FABRIC AND APPAREL · TOPIC 5 · LEVEL 3 TERM 2

1. From strength to feel

The tests of Topics 3 and 4 load a fabric to destruction. The properties in this sheet are measured at the small loads and deformations of ordinary handling — the loads a hand applies when it lifts, squeezes and strokes a fabric, and the loads gravity applies when a garment hangs. A fabric is bent, sheared, compressed and rubbed every time it is worn, and its response to those gentle actions is what the trade calls handle and what the wearer calls quality. These properties are governed by fibre stiffness, yarn structure, weave, thickness and finish, and they are strongly directional: warp-way and weft-way results are measured and reported separately.

All specimens are conditioned at 65% RH and 20 °C. Moisture changes fibre bending behaviour markedly — a fabric tested in a dry morning laboratory and a humid afternoon laboratory can appear to be two different fabrics — so handle measurements without conditioning are not repeatable.

2. Bending and the cantilever test

2.1 Principle — ASTM D1388 cantilever method

The cantilever test turns the fabric into its own measuring instrument. A rectangular strip of fabric is laid on a horizontal platform and slid slowly forward so that an increasing length overhangs the edge. The overhanging part bends under its own weight. When the tip of the overhang reaches a plane inclined at 41.5 degrees to the horizontal, placed below the platform edge, the overhang length is read. The angle is not arbitrary: at 41.5 degrees the geometry of the bending strip gives a particularly simple relation between overhang and the intrinsic bending property defined below.

$$\text{Bending length } c = \text{overhang length at } 41.5^\circ / 2$$

Bending length is a length, in centimetres, and it has a physical meaning: it is the length of a strip of the fabric that would bend under its own weight to a defined curvature. A limp fabric has a short bending length — a few centimetres or less. A stiff fabric has a long one. Because the measurement uses only the fabric's own weight as the load, it is non-destructive, rapid and well suited to routine comparison of warp and weft directions and of finishes.

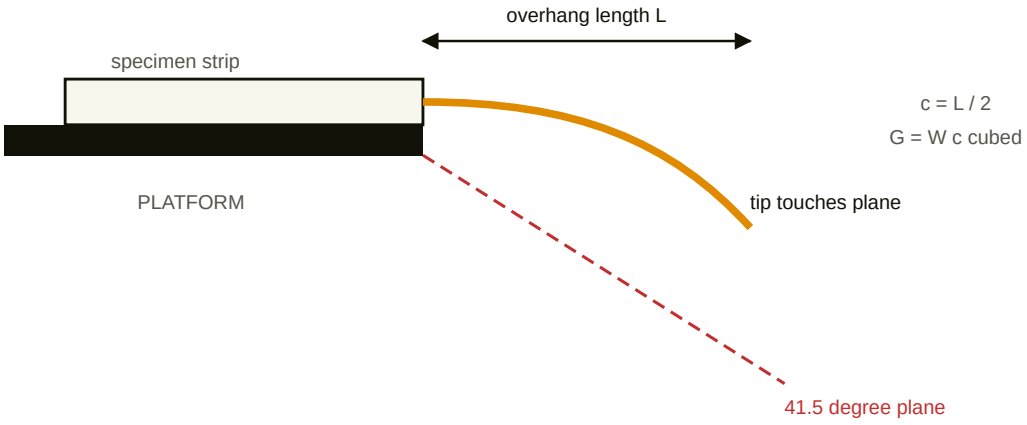


Figure 1 — Cantilever geometry. The strip slides forward until its tip meets the 41.5 degree plane; the bending length is half the overhang at that moment.

2.2 Flexural rigidity — Peirce relation

Bending length alone does not separate a light stiff fabric from a heavy limp one, because a heavy fabric bends more under its own greater weight. Peirce combined bending length with mass per unit area to give flexural rigidity, the fabric analogue of the engineer's bending stiffness:

$$G = W \times c^3$$

where G is flexural rigidity, W is the mass per unit area of the fabric, and c is the bending length, in consistent units. With W in mg/cm² and c in cm, G is obtained in mg·cm. Flexural rigidity rises with the cube of bending length, so a modest change in bending length is a large change in rigidity — a fabric whose bending length doubles is eight times as rigid at the same mass. Rigidity also rises steeply with thickness (Topic 1), because the outer fibres of a bent thick fabric are strained far more than those of a thin one.

WORKED EXAMPLE — BENDING LENGTH AND FLEXURAL RIGIDITY

Fabric mass per unit area = 185.5 g/m². Convert W: 185.5 g/m² = 185,500 mg / 10,000 cm² = **18.55 mg/cm²**.

Cantilever overhang at the 41.5 degree plane = 6.4 cm, so bending length c = 6.4 / 2 = **3.2 cm**.

$c^3 = 3.2 \times 3.2 \times 3.2 = 32.768$ cm³.

$G = W \times c^3 = 18.55 \times 32.768 =$ **607.8 mg·cm**, reported as about **608 mg·cm**.

> The cubic term dominates: had the same fabric finished to c = 4.0 cm, its rigidity would be 18.55 × 64 = 1,187 mg·cm — nearly double — from a bending-length change of only 0.8 cm.

2.3 Torsional stiffness — in outline

Bending describes a fabric flexing in one plane; twisting a strip describes its resistance to the torsional deformations that occur when a garment is wrung, folded diagonally or draped over a corner. In torsional tests a strip is suspended or clamped, twisted through a stated angle, and the restoring torque or the angle of twist under a stated torque is recorded. Torsional stiffness rises with bending rigidity and with inter-yarn friction, and it contributes to the springiness a hand feels when a fabric is crumpled. It is measured less routinely than bending, but it completes the picture of low-load stiffness and reappears in the instrumented handle systems of Topic 9.

3. Drape

3.1 What drape is

Drape is the way a fabric falls under gravity into folds when it is supported over only part of its area — the behaviour that decides how a skirt hangs, how a curtain folds and how a tablecloth lies over an edge. Drape is a combined response: it needs low bending resistance so folds can form, low shear resistance so the fabric can change shape to make those folds, and enough weight to pull the folds down. A fabric can be flexible in bending yet drape badly if it resists shear, which is why drape is measured directly and not calculated from bending length alone.

3.2 The Cusick drape meter

In the Cusick drape test a circular specimen is placed concentrically on a smaller horizontal support disc and allowed to drape down around it. Light projected from above (in the original instrument) or an image captured from above gives the projected area of the draped specimen — the shadow or plan-view outline of its folds. The projected area lies between two limits: the area of the support disc, if the fabric were infinitely stiff and stayed flat, projected as the full specimen area — treated carefully below — and the specimen's own area. The drape coefficient normalises the measurement between those limits:

$$\text{Drape coefficient } F = (A_d - A_2) / (A_1 - A_2)$$

where A_1 is the area of the flat circular specimen, A_2 is the area of the support disc, and A_d is the projected area of the draped specimen. F = 0 would mean the fabric collapses completely onto the disc outline; F = 1 would mean it does not drape at all and its projection remains the full specimen. Soft, fluid fabrics give low F; stiff fabrics give high F.

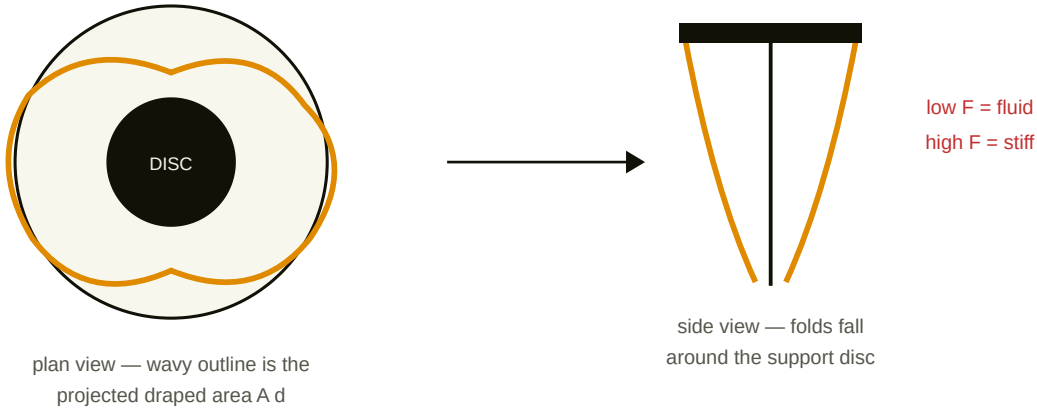


Figure 2 — Drape measurement. A circular specimen drapes around a smaller support disc; its projected area, between the disc area and the full specimen area, gives the drape coefficient.

WORKED EXAMPLE — DRAPE COEFFICIENT

Specimen radius 15 cm: $A_1 = \pi \times 15^2 =$ **706.9 cm²**.

Support disc radius 9 cm: $A_2 = \pi \times 9^2 =$ **254.5 cm²**.

Projected draped area $A_d = 480$ cm².

$F = (480 - 254.5) / (706.9 - 254.5) = 225.5 / 452.4 =$ **0.50** (0.498 before rounding).

The fabric sits midway between the two limits — a medium drape, suitable where some body is wanted in the folds. A fluid dress fabric would be expected well below this value and a stiff furnishing fabric well above it, but the acceptable band is always fixed by the end use, not by the instrument.

The number of fold nodes in the drape image is reported alongside F in careful work: two specimens of equal coefficient can drape into few deep folds or many shallow ones, and they do not look or handle alike.

4. Handle — the sum the hand perceives

Handle (or hand) is the total sensation of handling a fabric. No single instrument reads it, because the hand performs several tests at once:

- **Stiffness and spring** — bending and torsional response, measured above; the hand bends the fabric and feels it push back or collapse.
- **Compressibility and fullness** — the change of thickness under thumb pressure (Topic 1 thickness at low and higher pressures); a fabric that compresses softly feels full and warm, one that does not feels boardy or thin.

- **Surface properties** — smoothness, roughness, friction and hairiness as fingers stroke the surface; the same fabric can feel sleek in one direction and rough in the other.
- **Extensibility and recovery** — the small stretch and its return when the fabric is pulled gently between the fingers.
- **Thermal feel** — the initial coolness or warmth on contact, set by fibre conductivity and surface hair.

Because handle is multidimensional, the laboratory has two complementary routes. The first is disciplined subjective assessment: trained assessors rank specimens against reference standards for stated attributes (softness, stiffness, smoothness) under controlled conditions — still the reference method for product development, and valid only when assessors, references and vocabulary are controlled. The second is instrumented measurement of the component properties — bending, shear, compression, surface friction and roughness, tensile at low load — which is developed fully in Topic 9 through the FAST and Kawabata systems. The bridge is this: every subjective handle word corresponds to a combination of measurable low-load properties, and the work of Topics 5 and 9 together is to make that correspondence quantitative.

HANDLE WORD	MAIN MEASURED CONTRIBUTORS	TYPICAL DIRECTION OF CHANGE
Stiff / boardy	High G, high F	Rises with thickness, resin finish, tight weave
Limp / fluid	Low G, low F	Rises with soft fibres, loose weave, softener
Full / lofty	High compressible thickness	Rises with bulked yarns and raised finishes
Sleek	Low surface friction and roughness	Rises with filament yarns, calendering, singeing
Springy	High bending and torsional recovery	Rises with wool content and resilient finishes

5. Controlling stiffness, handle and drape

The designer and finisher move these properties with the same levers that moved strength, but often in the opposite direction. Increasing thickness, density or resin crosslinking raises stiffness and the drape coefficient while lowering softness; softeners, raising, and looser constructions do the reverse and usually cost tear strength (Topic 4). Fibre choice sets the starting point — fine, flexible fibres and filament yarns drape more readily than coarse, stiff ones at equal construction. The specification task is therefore a balance stated in numbers: a bending length or rigidity band for each direction, a drape coefficient band for hanging end uses, and handle references for the qualities instruments still assess comparatively — each measured on the finished fabric, in the standard atmosphere, in both directions.

Glossary

Bending length (c) — half the overhang at which a cantilever strip tip reaches the 41.5 degree plane. **Flexural rigidity (G)** — Peirce stiffness measure, G = W c^3. **Drape** — manner in which a fabric falls into folds under gravity when partly supported. **Drape coefficient (F)** — normalised projected draped area between the disc area and the specimen area. **Cusick drape meter** — instrument that projects or images the draped specimen outline. **Handle** — total tactile sensation of a fabric, the sum of stiffness, compressibility, surface and related properties. **Torsional stiffness** — resistance of a fabric strip to twisting. **Node** — one fold wave in a draped specimen outline.

Key points

• Cantilever bending (ASTM D1388): bending length c is half the overhang at the 41.5 degree plane.
• Peirce flexural rigidity G = W × c^3, with W as mass per unit area; rigidity is cubic in bending length.
• Worked fabric: W = 18.55 mg/cm^2, c = 3.2 cm, G = 607.8 mg·cm, about 608 mg·cm.
• Drape coefficient F = (A_d − A_2)/(A_1 − A_2); low F is fluid, high F is stiff; worked value F = 0.50.
• Handle is multidimensional — stiffness, compressibility and surface properties together — and leads directly to the FAST and Kawabata systems of Topic 9.

Practice

1. A fabric of 120 g/m^2 gives a cantilever overhang of 5.0 cm. Convert W to mg/cm^2, find c, and calculate G in mg·cm.
2. The fabric of Question 1 is softened and its overhang falls to 4.0 cm at the same mass. Calculate the new G and the percentage reduction in rigidity.
3. A drape specimen of radius 15 cm (A_1 = 706.9 cm^2) on a disc of radius 9 cm (A_2 = 254.5 cm^2) projects a draped area of 390 cm^2. Calculate F and state whether it drapes more or less fluidly than the worked example in this sheet.
4. Explain why two fabrics with the same bending length can drape differently, naming the additional property involved.

Answer key

1. W = 120 g/m^2 = 120,000 mg / 10,000 cm^2 = **12.0 mg/cm^2**. c = 5.0 / 2 = **2.5 cm**. c^3 = 15.625. G = 12.0 × 15.625 = **187.5 mg·cm**.
2. c = 4.0 / 2 = 2.0 cm. c^3 = 8.0. G = 12.0 × 8.0 = **96.0 mg·cm**. Reduction = (187.5 − 96.0) / 187.5 × 100 = 91.5 / 187.5 × 100 = **48.8%** — nearly half the rigidity lost from a 1 cm change of overhang, because of the cubic term.

3. F = (390 − 254.5) / (706.9 − 254.5) = 135.5 / 452.4 = **0.30** (0.299 before rounding). This is well below the worked example value of 0.50, so the fabric drapes **more fluidly**.
4. Bending length measures resistance to bending in one direction only. Drape also requires the fabric to shear — to let warp and weft rotate relative to each other — so that it can change shape and form folds. A fabric with low bending resistance but high shear resistance keeps its shape and drapes poorly despite a short bending length.

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